Symmetry-constrained spectral fibres: reverse-complement group actions on double-stranded DNA and the fibres of circular molecular fingerprints

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Abstract

A representation that depends on an object only through its local content is constant on the *spectral fibre* of the object, the set of objects with the same multiset of local pieces. For protein sequences the pieces are k-mers and the fibres are Eulerian trails of a de Bruijn graph. This paper treats two settings in which a symmetry enters. For double-stranded DNA the group $G = \{e, \rho\}$ generated by reverse complementation acts on sequence space, a strand-agnostic encoder is G-invariant, and its fibre is the set $\mathrm{DS}(S, k)$ of sequences with the same double-stranded spectrum. This fibre is not a single Eulerian count: it is a sum of BEST determinants over the points of a lattice, the (-1)-eigenlattice of ρ on the circulations of a double cover, and counting it is #P-hard. For bacteriophage ϕX174 at $k = 10$, $|\mathrm{DS}| = 31\,925\,753\,246\,212$, between the single-strand count $8\,610\,708\,632$ and the $1.4 \cdot 10^{36}$ Eulerian circuits of the double cover, which is what a naive count would give. For small molecules the local pieces are the atom environments of circular fingerprints, and the symmetry is built into their construction, because the neighbours of an atom are unordered. We prove that on substituted linear chains the radius-r Morgan (ECFP_{2r}) partition is exactly the spectrum of $(2r+1)$-windows read up to reversal, with the substituent at distance r coarsened to its atom type, and confirm this by exhaustive enumeration of 19 683 chains. We construct distinct molecules with identical ECFP4 and ECFP6 fingerprints (2048-bit, count and unhashed) in which the topological distance between two pharmacophoric groups changes from 7 to 12 bonds, and we measure that 3.2% of the 9 963 chains with nine substituted positions share their ECFP4 with another molecule. A common quotient framework covers composition encoders, convolutional encoders, strand-agnostic DNA encoders and circular fingerprints.

Keywords. reverse complement, double-stranded DNA, de Bruijn graph, BEST theorem, lattice sum, ϕX174, extended-connectivity fingerprint, Morgan algorithm, Weisfeiler–Lehman, quotient space, group action.

1 Introduction

A *local* representation of a discrete object depends on it only through the multiset of its local pieces. For a sequence the pieces are the k-mers; for a molecular graph they are the atom environments of radius r. Two objects with the same multiset receive the same representation, whatever the representation was trained to do. We called the set of such objects the spectral fibre of an object, computed its size exactly for every human protein, and showed which encoder architectures are confined to it [3, 4, 5, 6].

Those papers concern one strand of a linear sequence. Two settings that matter in practice carry a symmetry in addition. A regulatory element or a CRISPR target has no preferred strand, so a DNA encoder should give the same output to a sequence and its reverse complement. A molecule has no preferred orientation, and the environments that circular fingerprints hash are unordered. In both cases the fibre of the representation is larger than the one-strand fibre, and in the DNA case it is also much harder to count. The combinatorics of the double-stranded fibre is developed in the monograph [2]: the lattice-sum formula, the #P-hardness, the moves that connect the fibre, and the computation for ϕX174. The present paper does not add to that combinatorics. Its contributions are the following.

1. A formulation of both settings as quotients of an object space by a group action followed by a local statistic (Sections 2 and 5), which makes precise why a strand-agnostic encoder is blind to the double-stranded fibre and not only to the one-strand fibre.

2. An account of the ϕX174 numbers in this light (Section 3), including the comparison between the true fibre, the one-strand fibres and the Eulerian circuits of the double cover, which differ by 26 orders of magnitude.

3. An exact description of the fibres of Morgan fingerprints on substituted linear chains (Theorem 5), verified exhaustively; explicit collision pairs at ECFP4 and ECFP6 with changed pharmacophore geometry; and a census of collisions (Section 4).

2 Double-stranded DNA as a group quotient

Let $\Sigma = \{\mathtt{A}, \mathtt{C}, \mathtt{G}, \mathtt{T}\}$ and let ρ be reverse complementation, $\rho(S) = \overline{S}$. It is an involution on Σ^L and on circular sequences, so $G = \{e, \rho\} \cong \mathbb{Z}/2$ acts on sequence space. A *strand-agnostic* encoder is a G-invariant map ϕ, i.e. $\phi(S) = \phi(\rho S)$. It factors through the orbit space Σ^L/G.

Definition 1 (double-stranded spectrum)**.** The *double-stranded k-spectrum* of S is the multiset $\mathrm{spec}_k(S) + \mathrm{spec}_k(\rho S)$. For circular S, the *double-stranded fibre* is

$$\mathrm{DS}(S,k) = \{T \text{ circular}: \ \mathrm{spec}_{k+1}(T) + \mathrm{spec}_{k+1}(\rho T) = \mathrm{spec}_{k+1}(S) + \mathrm{spec}_{k+1}(\rho S)\},$$

counted with T and ρT as distinct elements unless equal [2, Def. 7.3]. The index convention follows [2]: $\mathrm{DS}(S,k)$ is defined by $(k+1)$-mers, the k-mers being the vertices of the graph below.

Proposition 2 (what a strand-agnostic local encoder sees)**.** *Let ϕ be local of order $k+1$ on circular sequences, i.e. $\phi(T)$ depends on T only through* $\mathrm{spec}_{k+1}(T)$*, and G-invariant. Then ϕ is constant on* $\mathrm{DS}(S,k)$*, and* $\mathrm{DS}(S,k) \supseteq F_{k+1}(S) \cup F_{k+1}(\rho S)$*.*

Proof. For $T \in \mathrm{DS}(S,k)$ the two strands of T together have the same $(k+1)$-mers as the two strands of S. A G-invariant ϕ may be written as $\phi(T) = \psi(\mathrm{spec}_{k+1}(T) + \mathrm{spec}_{k+1}(\rho T))$: indeed

$\phi(T) = \frac{1}{2}(\phi(T) + \phi(\rho T))$ and each summand is a function of one strand's spectrum, but the pair of spectra is determined only up to order by the sum; since $\phi(T) = \phi(\rho T)$, the value does not depend on which of the two spectra is called the first. The inclusion holds because $T \in F_{k+1}(S)$ gives $\mathrm{spec}_{k+1}(T) = \mathrm{spec}_{k+1}(S)$ and then $\mathrm{spec}_{k+1}(\rho T) = \mathrm{spec}_{k+1}(\rho S)$, the spectrum of the reverse complement being determined by that of the sequence. □

So the price of strand invariance is exact. A one-strand local encoder is blind to $F_{k+1}(S)$; a strand-agnostic one is blind to $\mathrm{DS}(S, k)$, which contains the one-strand fibres of both strands and, in general, much more.

The double cover and the lattice. Let D_k be the de Bruijn multigraph on the k-mers of S and of ρS together, with one arc per $(k+1)$-mer occurrence on either strand. Reverse complementation acts on D_k as a fixed-point-free involution ρ of the arcs that reverses direction [2, Ch. 5–6]. A circular sequence T with the double-stranded spectrum of S corresponds to a pair of Eulerian circuits, one through each strand, whose arc-usage vectors c and ρc add up to the multiplicity vector m of D_k. The usage c is a circulation of D_k, and $c + \rho c = m$ says that $f = c - \rho c$ lies in the (-1)-eigenlattice Λ^- of ρ on the circulation lattice, inside the box $|f| \le m$ with $f \equiv m \bmod 2$.

Theorem 3 (the fibre is a lattice sum [2, Thm. 7.6])**.**

$$|\mathrm{DS}(S, k)| = \sum_{f \in \Lambda^- \cap \{|f| \le m,\ f \equiv m\ (2)\}} N_{\mathrm{seq}}\Big(\tfrac{m+f}{2}\Big),$$

where $N_{\mathrm{seq}}(c)$ is the number of circular sequences that use each arc exactly $c(a)$ times, which is 0 *unless the support of c is connected and otherwise $t(G_c)\,\mathrm{Loc}(G_c)/\prod_a c(a)!$ by the BEST theorem [1]. The rank of Λ^- is $(|A| - |V| + 2 - \#\mathrm{fix}_A - \#\mathrm{fix}_V)/2$, the Hopf trace of ρ [2, Ch. 7, check L].*

Two consequences separate this fibre from the one-strand case. First, there is no single determinant: the sum runs over a lattice whose rank grows with the number of shared k-mers between the strands, and evaluating it is #P-hard in general [2, Cor. 10.6]. Second, the Eulerian circuits of D_k itself do not count the fibre. A circuit of D_k walks through both strands as one closed walk; it corresponds to a double-stranded reconstruction only when its usage vector is of the form c with $c + \rho c = m$, and each reconstruction is then counted with the multiplicity $\prod c(a)!$ and once per rotation. The BEST count of D_k is therefore an upper bound that is astronomically loose (Section 3).

Moves. On one strand, fibre members differ by transpositions of segments between repeated k-mers [12, 10]. On two strands there is a second move: a segment flanked by an inverted repeat $w \cdots \rho w$ may be replaced by its reverse complement without changing the double-stranded spectrum [2, Thm. 11.1]. Transpositions between direct repeats and inversions between inverted repeats are the two products of non-allelic homologous recombination, so a strand-agnostic local encoder is blind to exactly the rearrangements that recombination between repeats of length k produces. Whether the two moves together with ρ connect every double-stranded fibre is conjectured, and proved for the palindromic genomes of [2, Ch. 10–11].

3 The reconstruction ambiguity of ϕX174

Bacteriophage ϕX174 (RefSeq NC_001422.1) is a circular single-stranded DNA genome of 5 386 bp whose replicative form is double-stranded. Table 1 collects the numbers computed in [2] from the

k	rank Λ^-	box points	connected	$\lvert\mathrm{DS}(S,k)\rvert$	$N_{\mathrm{seq}}(S)+N_{\mathrm{seq}}(\rho S)$	$\mathrm{ec}(D_k)$
12	4	10	8	10	4	33
11	12	528	218	864	66	25 318 912
10	48	2 481 687 900	909 103 210	31 925 753 246 212	8 610 708 632	$1.368 \cdot 10^{36}$

Table 1: Double-stranded fibres of ϕX174 [2, Tables of Ch. 7–9]. Columns: rank of the (-1)-eigenlattice; lattice points in the box; points with connected support; the fibre; the sum of the one-strand fibres of the two strands; Eulerian circuits of the double cover.

lattice sum, by three independent algorithms at $k = 11$ and 12 (brute force, a fermionic frontier sweep and an arborescence frontier) and by a compiled enumeration of the half-box at $k = 10$, whose 1 240 843 950 points were processed modulo five primes and recombined by the Chinese remainder theorem [2, Ch. 8–9].

Three readings of the table. At $k = 12$ the two strands share one inverted-repeat pair and one palindrome (`TGGAGGCCTCCA`), the lattice has rank 4, and the fibre has 10 members against 4 for the two one-strand fibres. At $k = 10$ the fibre has $3.19 \cdot 10^{13}$ members. A strand-agnostic local encoder of order 11 assigns all of them one vector; a one-strand encoder of the same order is blind to $8.6 \cdot 10^{9}$ of them. The gap is the price of strand invariance for this genome. Finally, the double cover has $1.37 \cdot 10^{36}$ Eulerian circuits at $k = 10$. Anyone who counted the fibre by applying the BEST theorem to the double cover would overestimate it by a factor of $4 \cdot 10^{22}$. The overcount comes from circuits whose usage is not of the form $c + \rho c = m$ and from the labelling of parallel arcs.

The fibre is connected by the two moves of Section 2: at $k = 11$ all 864 reconstructions and at $k = 12$ all 10 are reached from S by inversion flips alone, in at most 8 and 3 flips [2, Ch. 7, `ds_flips`]. The reconstructions are not exotic: each is ϕX174 with some segments between inverted repeats reversed, exactly the sequences that an inversion event would produce.

4 Fibres of circular molecular fingerprints

4.1 Morgan identifiers as a symmetric local statistic

Extended-connectivity fingerprints [11] assign to each atom an identifier of its radius-r environment by r rounds of the Morgan algorithm [9]: the identifier of an atom at round i is a hash of its identifier at round $i-1$ and the unordered multiset of (bond type, neighbour identifier at round $i-1$). This is Weisfeiler–Lehman colour refinement [13, 14]. The fingerprint is the multiset of identifiers of all atoms at all rounds $0, \ldots, r$, with environments that cover an atom set already covered removed, hashed to a bit vector (ECFP_{2r}) or kept as counts. A molecular encoder built on ECFP is therefore local in the graph sense, and its fibre is the set of molecules with the same multiset of environments. The symmetry is intrinsic: because the neighbours are unordered, an environment is read up to the automorphisms that fix its centre.

4.2 Substituted chains: an exact description

Consider the molecules $\mathrm{CH_3}$-$(\mathrm{CHR_1})$-$\cdots$-(CHR_N)-$\mathrm{CH_3}$ with $R_i \in \{\mathrm{H}, \mathrm{F}, \mathrm{Cl}\}$. Each is a string over the alphabet $\{\mathtt{H}, \mathtt{F}, \mathtt{L}\}$ (for $\mathrm{CH_2}$, CHF, CHCl), bounded by terminal methyls `M`, and two strings give the same molecule if and only if they are equal or mirror images. For a letter x let $\bar{x}$ be its *atom type*: $\bar{\mathtt{F}} = \bar{\mathtt{L}} = \mathtt{X}$ (a carbon of degree 3 with one hydrogen), and $\bar{\mathtt{H}} = \mathtt{H}$, $\bar{\mathtt{M}} = \mathtt{M}$.

fingerprint	classes	classes with ≥ 2 molecules	molecules in such classes	largest class
ECFP2 ($r = 1$)	2 729	1 910	9 144 (91.8%)	60
ECFP4 ($r = 2$)	9 803	158	318 (3.2%)	3
ECFP6 ($r = 3$)	9 963	0	0	–

Table 2: Collisions of unhashed Morgan count fingerprints among the 9 963 chain molecules $CH_3(CHR)_9CH_3$, $R \in \{H, F, Cl\}$.

Definition 4 (coarsened symmetric window spectrum)**.** For the chain string $s = \mathtt{M}s_1 \cdots s_N\mathtt{M}$ and $r \geq 1$, the radius-r window of position i is $w_r(i) = (\bar{s}_{i-r}, s_{i-r+1}, \ldots, s_{i+r-1}, \bar{s}_{i+r})$, with the symbol $\wedge$ for positions outside the chain. The *symmetric window spectrum* $\Omega_r(s)$ is the multiset of the classes $\{w_r(i), w_r(i)^{\mathrm{rev}}\}$ over all positions i.

Theorem 5 (fibres of Morgan fingerprints on chains)**.** *Two chain molecules have the same unhashed radius-r Morgan count fingerprint if and only if $\Omega_r(s) = \Omega_r(s')$. Consequently the $ECFP_{2r}$ fibre of a chain is the set of chains with the same symmetric window spectrum: the $(2r + 1)$-window spectrum read up to reversal, in which the substituent at distance exactly r is seen only through its atom type.*

Proof. Write $c_i(v)$ for the identifier of atom v after i rounds. Round 0 gives the atom type. By induction on i, the identifier c_i of a chain carbon at position p is an injective function of the window $(\bar{s}_{p-i}, s_{p-i+1}, \ldots, s_{p+i-1}, \bar{s}_{p+i})$ read up to reversal: the induction step combines $c_{i-1}(p)$ with the unordered pair $\{c_{i-1}(p-1), c_{i-1}(p+1)\}$ (and, for a substituted carbon, the identifier of its halogen, which is determined by $c_{i-1}(p)$ and the halogen type). The pair is unordered, hence the reversal; the letters at distance i enter only through c_0, hence the coarsening; and when $s_{p-1} = s_{p+1}$ the two possible assignments of the outer letters give windows that are reversals of each other, so no information is lost. The identifiers of the halogen atoms and of the terminal methyls at round i are functions of their carbon neighbour's identifier at round $i - 1$, hence of windows already counted, and the identifiers at rounds below r are coarsenings of those at round r. Removing duplicate environments is a deterministic function of the graph structure that is the same for all chains. Hence the multiset of all identifiers is determined by $\Omega_r(s)$, and conversely determines it because the round-r identifiers of the chain carbons are injective on window classes. □

We verified the theorem by enumerating all $3^9 = 19\,683$ strings with $N = 9$ (9 963 distinct molecules), computing the unhashed count fingerprint with RDKit [8] for $r = 1, 2, 3$, and comparing the resulting partition with the partition by Ω_r: they coincide exactly at all three radii. The census in Table 2 follows.

The ECFP4 collisions are copy ambiguities of the kind found for proteins in [7]. For instance, the three chains

`CC(F)C(F)C(F)C(Cl)C(F)C(F)C(F)C(F)C(C)Cl`, `CC(F)C(F)C(F)C(F)C(Cl)C(F)C(F)C(F)C(C)Cl`, `CC(F)C(F)C(F)C(F)C(F)C(Cl)C(F)C(F)C(C)Cl`

form one class. The chlorine sits at position 4, 5 or 6 of a run of nine substituted carbons, and a radius-2 environment cannot tell which.

4.3 Explicit collision pairs and pharmacophore geometry

Proposition 6 (transposition between repeated linkers)**.** *Let A-u-X-u-Y-u-Z-u-B be a chain in which $u = (CH_2)_\ell$ and X, Y, Z are distinct substituted carbons, and let A-u-Y-u-X-u-Z-u-B be*

pair ($X = \mathrm{CHCH_3}$, $Y = \mathrm{CHCl}$, $Z = \mathrm{CHOH}$)	ECFP2	ECFP4	ECFP6	ECFP8
linear, $\ell = 4$	$=$	$=$	$\neq$	$\neq$
linear, $\ell = 5$	$=$	$=$	$=$	$\neq$
linear, $\ell = 6$	$=$	$=$	$=$	$\neq$
macrocycle, $X\,Y\,Z\,W$ vs $X\,Z\,Y\,W$, $\ell = 4$, $W = \mathrm{CHBr}$	$=$	$=$	$=$	$\neq$

Table 3: Collision pairs checked with RDKit. "=" means that the 2048-bit fingerprint, the 2048-bit count fingerprint and the unhashed identifier multiset all coincide; "≠" that all three differ. The linear pairs exchange X and Y; the macrocycle pairs exchange the cyclic order of two of four substituents. The bound $\ell \geq 2r - 1$ of Proposition 6 is attained: $\ell = 4$ separates at $r = 3$ and $\ell = 6$ at $r = 4$.

the chain with X and Y exchanged. The two molecules are distinct, and their radius-r Morgan fingerprints coincide if $\ell \geq 2r - 1$.

Proof. Distinctness: the reversed second chain is B-u-Z-u-X-u-Y-u-A, which differs from the first in the position of Z. Invariance: by Theorem 5 it suffices that the symmetric window spectra agree. A window of radius r that contains X or Y at its centre or in its interior sees full letters only within distance $r - 1$, and $\ell \geq 2r - 1$ places the next substituent at distance at least $2r$ from X; so every window meeting X lies inside u-X-u and is unchanged by the exchange, and likewise for Y. Windows not meeting X or Y are unchanged trivially. □

Table 3 lists the pairs we checked. The linker-4 pair,

`CCCCCC(C)CCCCC(Cl)CCCCC(O)CCCCC` and `CCCCCC(Cl)CCCCC(C)CCCCC(O)CCCCC`,

has identical ECFP4 in all three forms, and the two molecules place their pharmacophoric groups differently: the shortest-path distance from the chlorine to the hydroxyl is 7 bonds in the first and 12 in the second, while the methyl-to-chlorine distance is 7 in both. A model that predicts a property from ECFP4 assigns the two molecules the same value. Whether the property differs is a question about the property; the representation cannot express the difference. A macrocycle with four substituents in two cyclic orders gives the same at ECFP6 (Table 3). These examples are the exact analogue of the transposition move on sequences, with the linker u playing the role of the repeated $(k - 1)$-mer.

4.4 Stereochemistry

The default Morgan fingerprint carries no stereochemistry. In RDKit, L- and D-alanine have identical ECFP4, and so do (E)- and (Z)-2-butene; with the chirality option both pairs are separated. Enantiomers and geometric isomers are thus fibres of the default fingerprint that a single flag removes. The collisions of Table 3 are not of this kind: no option of the algorithm separates them at the stated radius, because the environments are identical as graphs.

5 The unified quotient

The four settings share one structure. Let $\mathcal{X}$ be a set of objects (strings, circular strings, graphs), let G be a group acting on $\mathcal{X}$ (trivial, $\{e, \rho\}$, or the identity when the symmetry is already inside the statistic), and let $\sigma_r \colon \mathcal{X} \to \mathcal{M}$ be a *local statistic* of radius r, a map to multisets of local

encoder	objects, G	local statistic σ	fibre and its size
composition, PseAAC ($\lambda \leq k-1$), CNN of window k with multiset pooling	strings, trivial	k-mer multiset (and first $(k-1)$-mer)	$F_k(S)$; one BEST determinant [3]
strand-agnostic DNA encoder of order $k+1$	circular strings, $\{e, \rho\}$	double-stranded $(k+1)$-mer multiset	$\mathrm{DS}(S,k)$; lattice sum of BEST determinants, #P-hard [2]
ECFP_{2r}, Morgan count fingerprints	graphs, automorphisms of environments	multiset of radius-r WL identifiers	on chains: symmetric coarsened window spectrum (Thm. 5); in general the WL-r class

Table 4: The quotient framework across the four encoder families.

pieces that is constant on G-orbits. The *quotient map* is $\pi = \sigma_r \colon \mathcal{X} \to \mathcal{M}$, and the *fibre* of x is $\mathcal{F}(x) = \pi^{-1}(\pi(x))$.

Proposition 7. *Let ϕ be any encoder that factors as $\phi = g \circ \sigma_r$. Then (i) ϕ is G-invariant; (ii) ϕ is constant on $\mathcal{F}(x)$ for every x; (iii) $\mathcal{F}(x)$ is a union of G-orbits and contains the orbit of x; and (iv) if $\sigma'_{r'}$ refines σ_r (i.e. $\sigma_r = h \circ \sigma'_{r'}$), then the fibres of $\sigma'_{r'}$ are contained in those of σ_r.*

Proof. All four are immediate from the factorization and from the G-invariance of σ_r. □

Table 4 instantiates the framework. What differs between the rows is how hard the fibre is to count and how large it is. On one strand the fibre is a determinant. With a free involution on the arcs it becomes a lattice sum whose evaluation is #P-hard. On graphs the fibre is the class of the r-round Weisfeiler–Lehman refinement, whose enumeration is not known to be tractable at all, but whose members can be constructed by the transposition principle wherever the graph contains repeated linkers. The framework also says what the escapes are, in every row alike: an encoder leaves the fibre only by using a statistic that is not a function of σ_r, e.g. a larger radius, absolute position, or, for molecules, three-dimensional coordinates or stereo flags.

6 Discussion

Symmetry makes fibres larger and harder to count, but it does not change their nature. In both settings the fibre is generated by moves that exchange interchangeable pieces across repeated boundaries: segments between repeated k-mers on one strand, additionally segments between inverted repeats on two strands, and blocks between repeated linkers in a molecular graph. In each case the move preserves the local content exactly, and in each case it changes something a downstream task may depend on: the gene order of a genome, or the distance between two pharmacophores.

Two limitations. The exact description of Morgan fibres (Theorem 5) is for substituted chains; for general graphs we have only the WL characterization and explicit constructions. And we did not measure any molecular property on the collision pairs: whether the linker-4 pair differs in a bioassay is an experimental question. What is settled is that no model reading ECFP4 can predict such a difference.

Data and code availability

`code/ecfp_fibres.py` (Python, RDKit 2026.03.6) reproduces Section 4: the collision pairs and their fingerprints at radii 1–4, the pharmacophore distances, the exhaustive check of Theorem 5 for $N = 9$ and the census, and the stereochemical examples; its transcript is `results/ecfp_fibres_output.txt`. The ϕX174 numbers of Section 3 are those of the verification scripts of the monograph's repository (`ds_lattice.py`, `ds_flips.py`, `ds_fermion.py`, `ds_count.py` at `https://github.com/Ruqing1963/algebraic-genomics`), whose transcripts we quote. The genome (NC_001422.1) is included. Code, data and this paper are at `https://github.com/Ruqing1963/symmetry-fibres`.

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